

Class Test -I (Nov. - 2022)
AMR2C1- Applied Mathematics -II
(B.E. 1yr ETC A, B)

Maximum Marks: 20

Time: 70 min.

Note: Attempt all four questions. Questions must be solved at one place. Each step should be clear and well defined.

- Q.1 If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ show that $A^2 - 4A - 5I = 0$ where $I, 0$ are the unit matrix and the null matrix of order 3 respectively. Use this result to find A^{-1} . 05
- Q.2 Examine whether the following system of equations is consistent. If so, find the solution: $x + 2y + 3z = 14$; $3x + y + 2z = 11$; $2x + 3y + z = 11$. 05
- Q.3 Define a differential equation with its types. Also form the differential equation by eliminating arbitrary constants:
(i) $y = A\cos x + B\sin x$. (ii) $y^2 = Ax^2 + Bx + C$. 05
- Q.4 Solve $(1 + 2xy\cos x^2 - 2xy)dx + (\sin x^2 - x^2)dy = 0$. 05